Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.2 – Distance–Time Graphs: Match the Graph

Unit 2: Forces and Motion

Warm-up

A person runs **250 m** in **30 seconds**. What is their speed? Give your answer to 1 decimal place.

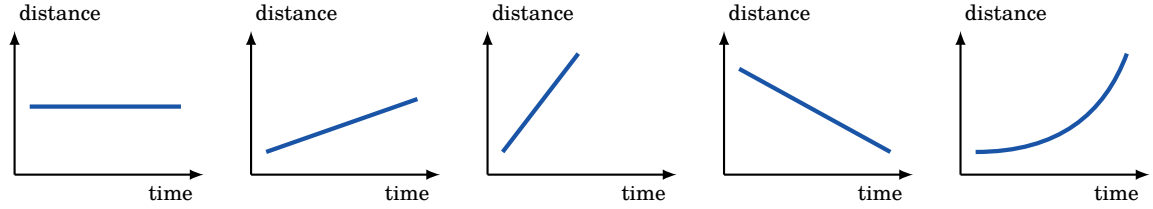
speed = distance ÷ time = **250** m ÷ **30** s = **8.3** m/s

Check: speed × time gives you back 250 m.

Reading a distance–time graph

Distance–time graph a graph with distance up the side and time along the bottom. Its shape tells the story of how something moved.

Today the distance is **your distance from a motion sensor**. Walk **away** from the sensor and the line goes **up**. Walk **towards** it and the line goes **down**. Stand still and the line stays **flat**.

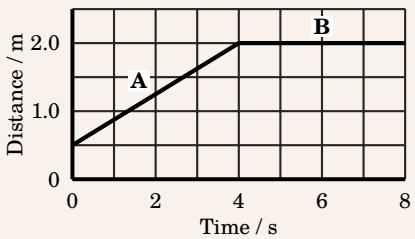


Stationary	**Constant speed**	**Faster**	**Constant speed**	**Speeding up**
standing still	away from it	steeper line	towards it	line curves up

One idea explains every shape: **the steepness of the line shows the speed.** Whatever the speed does, the steepness does too.

- A **flat line** has no steepness: no speed, so **stationary**.
- A **straight line** keeps the same steepness, so the speed is **constant**. The **steeper** the line, the **faster** the speed.
- A **curved line** changes steepness, so the speed is **changing**: getting steeper is speeding up, getting flatter is slowing down.
- **Up or down** shows the direction: up is **away from** the sensor, down is **towards** it.

Worked example — read the story



Part A is a straight line going up: the walker moved **away** from the sensor at a **constant speed**, from 0.5 m to 2.0 m in 4 s.

Part B is flat: the walker was **stationary**, standing 2.0 m from the sensor for the last 4 s.

Match the Graph — the method

Walk the graph

Kit: a PASCO motion sensor on a laptop running SPARKvue, pointing along a clear strip of floor. Your teacher sets it up so the screen shows a graph of **position** (your distance from the sensor, in metres) against **time**, on the same axes as the graphs overleaf.

Steps

1. **Plan first.** Read the graph, and decide where to start standing and how to move. Use the numbers: a line starting at 1.5 m means you start 1.5 m from the sensor.
2. **Walk it.** Stand at your starting distance, facing the sensor. On **Start**, move the way you planned. The line on the screen follows you.
3. **Compare.** Hold your line up against the graph on the sheet. Adjust the plan and walk it again until the two match.
4. **Write it down.** Under the graph, describe how you moved, using the words in the box below. Tick *Matched it?* when your line fits.

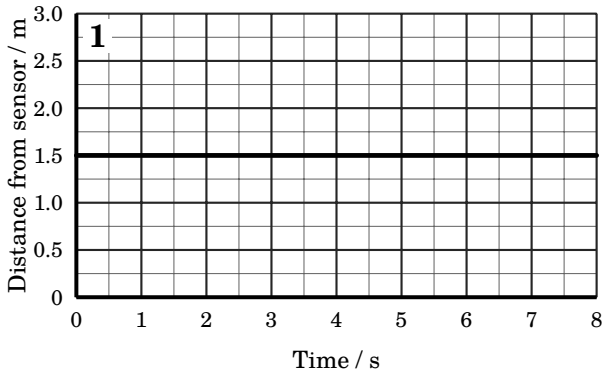
Check: walking away makes your line climb; walking towards makes it fall.

Care: keep the strip in front of the sensor clear of bags and chairs, and walk — the sensor reads the first thing in front of it.

Words to use: stationary • constant speed • away from • towards • faster • slower • speeding up • slowing down

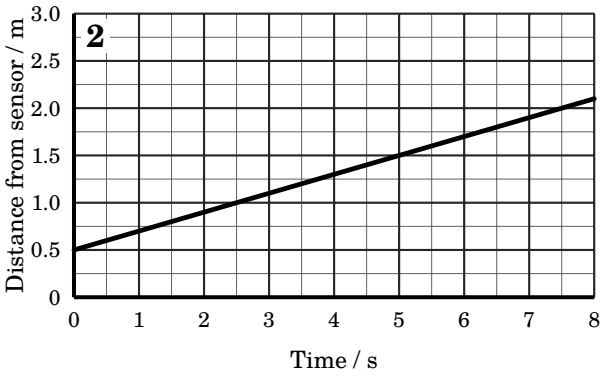
The six graphs

Work through them in order. Number 5 has four parts — describe each one.



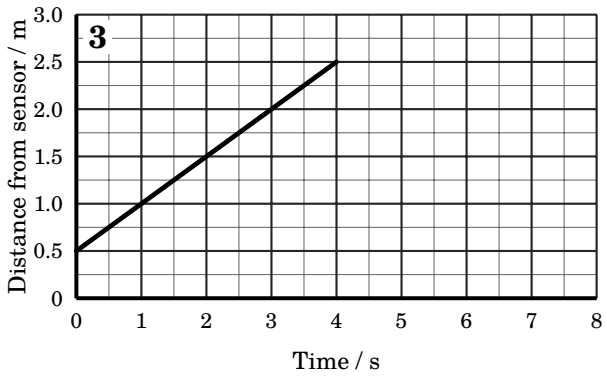
How I moved: **Matched it?** ☐

Stood still (stationary), 1.5 m from the sensor, for all 8 s.



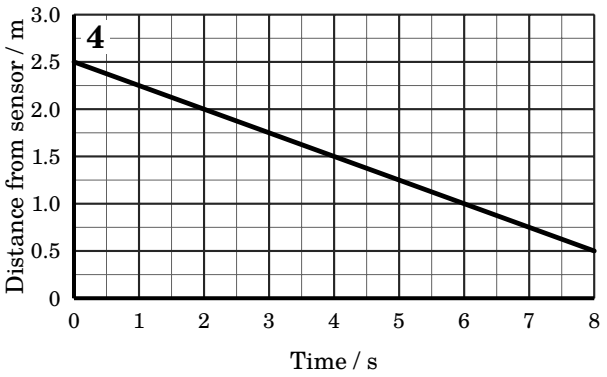
How I moved: **Matched it?** ☐

Walked away from the sensor slowly, at a constant speed, from 0.5 m to 2.1 m.



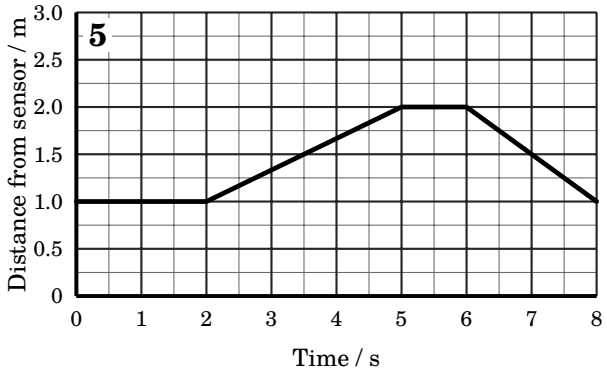
How I moved: **Matched it?** ☐

Walked away at a constant speed, faster than in 2: from 0.5 m to 2.5 m in only 4 s. The line is steeper.



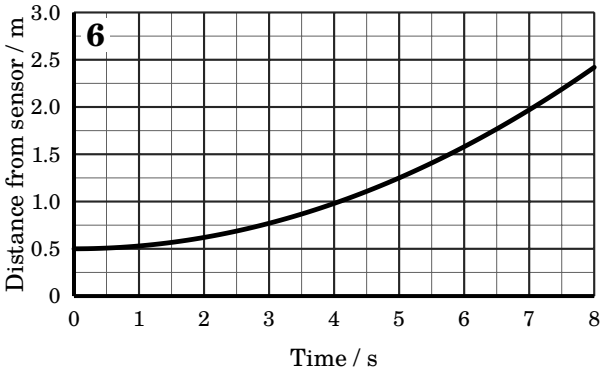
How I moved: **Matched it?** ☐

Walked towards the sensor at a constant speed, from 2.5 m to 0.5 m.



How I moved: **Matched it?** ☐

Stationary at 1.0 m for 2 s; walked away at a constant speed to 2.0 m; stationary for 1 s; walked back towards the sensor to 1.0 m, faster than on the way out.



How I moved: **Matched it?** ☐

Walked away from the sensor, speeding up: slow steps at first, getting faster.

What the graphs show

1. On a distance–time graph, a flat line means the walker is **stationary**.

2. Graphs 2 and 3 are both straight lines going up. Which one was the faster walk, and how can you tell from the graph?

Model answer: Graph 3, because its line is steeper: it covers more distance in each second.

3. Use Graph 3 to work out the walker's speed. They moved from 0.5 m to 2.5 m in 4 s.

speed = distance ÷ time = **2.0** m ÷ **4** s = **0.5** m/s

Check: the distance is how far they *moved*, end minus start.

To go further — the impossible graph

Nobody in the room can walk a graph that goes **straight up**, a vertical line. Explain why, using what a vertical line would mean about distance and time.

Model answer: A vertical line means the distance changes while no time passes at all. That would need an infinite speed, so no one can move from one place to another in zero time.